Quantitative Methods – Human Sciences

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Outline

Blocking backdoor paths

During this lecture, we will cover one of the most popular identification strategies: selection on observables, also known as conditional ignorability/independence (stats), unconfoundedness (econ), or conditional exchangeability (epi)

- First, we will locate it under the general umbrella of observational studies.
- Then we will review the required assumptions and show the identication results
- Finally, we will discuss common estimation methods: regression (including imputation), subclassification and matching, weighting, and a preview of augmented estimators

Observational studies

What is an observational study?

We already discussed the importance of **random assignment** to justify drawing causal conclusions and conduct statistical inference

Observational data is generated when the treatment of interest is **not** in control of the researcher (e.g., survey data, administrative records, etc.)

An observational study is an attempt to use observational data to *approximate* causal conclusions (or, as some would say, “to approximate an experiment”)

While, in experiments, (conditional) exchangeability is justified by design, in observational studies it relies on untestable and extra-statistical assumptions

Credible observational studies

Not all comparisons are created equal!

It is all too common to run some regression analysis and analyse its output as-if a credible *causal* ("controlled") comparison can be immediately obtained from the data

Most times, just recognizing that "our results are not causal" is not enough to prevent a causal interpretation!

A credible observational study should be carefully designed to address common issues and rule out alternative explanations

Credible observational studies

- Assignment is not random, but is it well understood?
- Can we explain why some units that look similar receive different treatments?
- Do we have reliable measures of our variables of interest?
- Does the timing of measurement allow us to order them?
- Can we check “balance” in observable characteristics?
- Can we quantify the influence that unobserved confounding would have on our results?
- Can we rule out alternative explanations to the observed “treatment effect”?

Unconfoundedness

General setting

Let's assume the following:

- We observe units $i = 1, \ldots, n$
- We observe a treatment
- We assume potential outcomes are well defined

Our quantity of interest can be (most commonly) the ATE or the ATT

We have a collection of **pre-treatment** covariates:

- Predetermined / causally precede
- If causes both and , it would *confound* the relationship between them
- If is potentially affected by , then it may possibly be a *mediator*, and it should be handlded with double care

More formally: is a valid adjustment set according to the backdoor criterion

Assumptions

The main assumption in this context is usually referred as unconfoudedness, conditional ignorability, or conditional exhangeability

It can be formalized as:

> **Read it like this**
>
> Among units with the same values of , the treatment is as-good-as randomly assigned

To actually be able to condition by , we also require positivity (or overlap): the treatment is not deterministic at any relevant value of

Identification of ATE/ATT

Intuition: Within strata of X, we have an experiment

Proof: Let's start with the strata effects

Then, let's use the **positivity** assumption:

Estimation

Estimation overview

We have seen that, under our non-parametric assumptions, the ATE/ATT is identified by the adjustment formula

In practice, there are several ways to implement these ideas, and different estimators incorporate further assumptions

The most widely used are

- Subclassification and Matching
- Weighting
- Regression
- Augmented estimators

Let’s take a look at each in general, before we move into a practical application

Modeling the treatment assignment

Subclassification

When is discrete, the subclassification estimator arises naturally, as seen before

We only need to estimate strata-specific effects ans average according to the reference population shares

Are we estimating any model? Are there “parameters” in the subclassification estimator?

As the number of covariates increases, however, the estimator becomes unfeasible

This is why other estimators have become much more widely used

Matching

For each treated unit with covariates , we will estimate

For treated units, we immediately have . But where do we get from?

- **Matching**: borrow the missing potential outcomes from control units with (nearly) the same

The estimator is then:

Where is the outcome of unit , which is closes to in terms of their covariates

In the simples case, is a single unit, without replacement. In more complicated versions, it can be an (weighted) average of units that are reused several times

> **But beware of the curse of dimensionality!**

Remember: Matching is NOT an identification strategy

Weighting

The idea of weighting comes from survey research: how do we get a representative sample?

It is similar, but kind of in the opposite direction, to the previous strategies:

- Instead of strata-specific effects that are then reweighted by , make look similar across groups and then take a simple average!

Recall that the difference-in-means can be viewed as:

How to solve the discrepancy in the weights?

We want the weights of the diff-in-means to be for everyone, so we can re-weight by (but, note that this requires a model for !)

What about propensity scores?

The weights just mentioned are usually referred as *(stabilized) propensity score weights*. Other names for the same object are *inverse probability of the treatment weighting* or simply *inverse probability weighting* (IPW)

Those weights require estimating a model for the treatment assignment . In other words, they require we fit a *propensity score* model, (or)

Using propensity score for weighting (or matching, or regression adjustment) is based on a simple but powerful result. If:

Then,

The propensity score is a *balancing score*, that will achieve balance in the variables used to estimate the propensity score, *in expectation*

Propensity score weights

For the ATE, the corresponding IPWs are:

- What would this weights do for the treated units? And for the controls?
- How would you modify these weights to target the ATT?

Hint: multiply by

Balance

Traditionally, balance in the distribution of covariates is assessed based on the absoluted standardized difference in means (ASMD):

Chattopadhyay, Hase, and Zubizarreta (2020) proposed using a *targeted* version of this instead:

Balance

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Which can be further refined as:

Where is a transformation of the original covariate.

Remember: Propensity score is NOT an identification strategy either!

Optimization (or calibration) weights

Usually, when using propensity score weights (or matching), what we really want to achieve is balance on covariates. So the process will imply:

1. estimate a model for ;
2. invert and and reweight sample;
3. check balance;
4. re-estimate propensity score if necessary.

Is that the only way?

Optimization (or calibration) weights

But we can also *bypass* these step, and directly search for weights that exactly (up to a tolerance level) “balance” (moments of) the covariates, through optimization approaches *calibration*!

If feasible, this will give you balance *in the sample*

Also, remember that balance is not the only desirable property: attention to variability vs stability, and positive vs negative weights (interpolation vs extrapolation)

Common ideas

Subclassification, matching and weighting share some common ideas

Most importantly, they all allow for some diagnostics *before* conducting the outcome analysis: balance testing

Checking balance does not require looking at the outcome, and it can provide a guidance on the credibility of the results to come

- This is what Rubin meant by “design triumphs analysis”

But be careful of overly relying on balance checking! (And use best practices, like equivalence testing)

Also, beware of residual imbalance! Modeling the outcome can come handy, as we will see later

Modeling the outcome expectation

Regression: look at *the* coefficient!

We can also take a model-based approach to estimate the causal effect of interest

For simplicity, let's assume that the potential outcomes follow certain functional form:

Which means that

and

What are we assuming about the treatment effect? What are we assuming about the potential outcomes? Are those assumptions sensible?

Regression: use it for imputation!

One way of making use of models that is less sensitive to the specification of a single coefficient is to use the model for imputation.

- Use the model(s) to predict the potential outcomes, and then calculate the difference of interest
- For inference, use the bootstrap

In epidemiology, this is known as g-computation (“g” stands for “generalized”)

This works incredibly well, above and beyond what looking at a single coefficient would give you

Even more, this can be paired with flexible machine learning approaches, going beyond linear regression models

Augmented estimators

Augmented IPW

Can we get the best of both worlds? Yes, we can!

Combine balancing (weighting) with outcome modeling (regression) in a single estimator: augmented IPW

Augmented IPW

This has the property of **doubly robustness** (DR): the model with be consistent for the treatment effect if either

1. the outcome model or,
2. the treatment model, are well specified

DR can also be interpreted as multiplicative errors approach: you will get closer to the truth (and *faster*) if you combine two not-perfect models than betting everyting to a single model.

> **Important**
>
> In terms of identification, the augmented estimator requires the same unconfoundedness assumption! No free lunch

What about machine learning?

There are different ways of using machine learning for estimation

- Flexible model of the outcome + imputation
- Flexible model of the treatment + weighting

Both have problems, so two frameworks have been proposed to incorporate ML methods in the estimation of treatment effects in a principled way:

- Targeted Minimum Loss Estimation (TMLE)
- Double/Debiased Machine Learning (DML)